Variable Phenomena and Logical Transductions

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Overview

(1) Traditionally, two types of phonological knowledge are of interest (*cf.* Anderson 1985, Heinz 2018).

   a. *Representational Knowledge*: what is a possible long term memory representation? what is a possible output of the phonological grammar?

   b. *Process Knowledge*: how do phonological inputs relate to phonological outputs?

(2) Some researchers believe that phonological knowledge should include the rate at which generalizations hold based on things like *The law of frequency matching*.

   a. Hayes et al. (2009: p. 826): "speakers of languages with variable lexical patterns respond stochastically when tested on such patterns. Their responses aggregately match the lexical frequencies."

   b. See also Magri (2023: p. 4): "If we do categorical phonology based on fieldwork data...there are no generalizations for us to capture here. There is nothing for us to do. If we instead turn from categorical data to frequencies extracted from corpora or obtained through psycholinguistic experiments, we will now see that straightforward phonological generalizations emerge."

(3) There is no denying that variable/gradient behavior exists, but the question always becomes, "how do we include gradience in our formal phonological models?"

   a. Option A: gradience goes directly into the phonological grammar (Hayes 2022)[1]

   b. Option B: gradience emerges outside the narrow phonological grammar ⇐

(4) Our goal as theorists should be to formally explain what the capacities are that underly human cognition (Cummins 1983) which can be defined as mathematical functions (van Rooij and Baggio 2021, Guest and Martin 2021, Nelson 2024).

   a. *Capacity 1*: our phonological system can generate more than one possible output for a given input.

   b. *Capacity 2*: specific outputs at any moment are selected based on a variety of different factors.

(5) In the remainder of the talk I will formalize the previous idea under the broad view of "model-theoretic phonology" which provides clarity in expressing the properties of different linguistic systems. In this instance, semirings and weighted logic provide a way to distinguish between *phonological knowledge* and *phonological usage*.

(6) Nasal place assimilation in English will be the phenomena that empirically grounds the idea (Borowsky 1986, Avery and Rice 1989, Coleman et al. 2016).

[1] See Appendix B for an overview.

Guiding Principles

(7) Probabilities should model ignorance, not actual behavior.

> *"...the reification of probabilities as cognitively represented quantities invites a problematic conception of randomness and probability, in which randomness plays a shallow role in the generation of behavior [...] The states of physical systems are not governed by probabilities. Probabilities are merely statistical descriptions that are useful when we are mostly ignorant about the detailed dynamics of a system and its surroundings."* - Tilsen (2023: p. 265)

(8) Individual grammars should model knowledge of language, not use of language.

> *"...it is perhaps worth while to reiterate that a generative grammar is not a model for a speaker or a hearer. It attempts to characterize in the most neutral possible terms the knowledge of the language that provides the basis for actual use of language by a speaker-hearer."* - Chomsky (1965: p. 9)

(9) Phonological knowledge is only one of many factors governing behavior.

> *"...linguistic knowledge is only one of the inputs to language production, language comprehension, and other forms of language performance. What accounts for the facts of performance is a conjunct of a theory of linguistic knowledge ('What is the nature of the representation of linguistic knowledge?') and a theory of language performance ('How is this knowledge put to use?')."* - Mohanan (1986: p. 183)

(10) The factors governing behavior are

> *"...an individual speaker's production of variability in any given instance of language use is shaped by three types of conditioning factors: sociostylistic ('s-conditioning'), internal linguistic ('i- conditioning'), and psychophysiological ('p-conditioning'). With respect to the third of these, one of our main lines of argument is that looking at the individual-level dynamics of variation requires a careful examination of general cognitive systems (for example, those related to memory) and psychophysiological systems (like those involved in articulation and perception)...Psychophysiological conditioning factors and internal linguistic conditioning factors are architecturally distinct."* - Tamminga et al. (2016: p. 301)

Model-Theoretic Phonology

(11) A **model signature** $\mathcal{M}$ is a collection of symbols for the functions, relations, and constants that describe structures.

a. This provides ingredients for formally defining phonological representations.

b. For example, the signature $\langle \lhd, \{R_\sigma \mid \sigma \in \Sigma\}\rangle$ gives us two types of operations. The *ordering relation* $\lhd$ lets us linearly order individual elements and the *labeling*

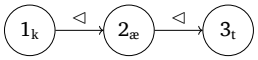
relations R_σ allows us to give a label to an element where the labels are drawn from a finite set Σ.

(12) An $\mathcal{M}$**-structure**, A, contains a set called the **domain**, as well as **denotations**

a. The domain is typically an initial sequence of the natural numbers: {1,2,…}.

b. A denotation takes the domain and says which elements of the domain satisfy which properties.

c. Given a set of symbols Σ = {æ,t,k}, the structure for the word *cat* is $\langle \mathcal{D} = \{1,2,3\}; \lhd = \{(1,2),(2,3)\}; \{R_{\text{æ}} = 2, R_{\text{t}} = 3, R_{\text{k}} = 1\}\rangle$.

d. We can also display this graphically: 

(13) A **logical language** in logic x is defined by combining the symbols of that logic with a specific model signature $\mathcal{S}$. Here, I will assume **Monadic Second Order** (MSO) logic and its weighted variant.

a. The model signature provides the tools to formalize representations.

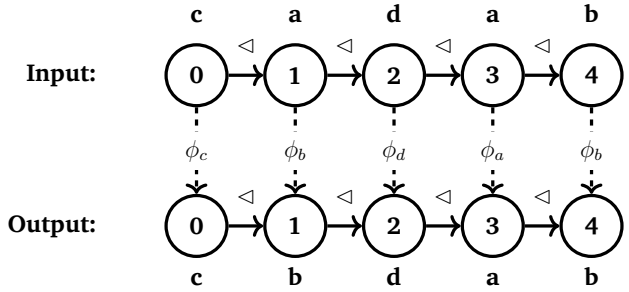
b. The logic provides a way to do inference over the representations that are built.

(14) An **interpretation** of structure A in terms of structure B is a function denoted by a set of n formulas $\{\phi_i, \ldots, \phi_n\}$ where n is equal to the number of functions, relations, and constants in A's model signature (*cf. logical transductions:* Courcelle 1994, Engelfriet and Hoogeboom 2001).[2]

a. A formula $\phi_P(x) := Q(x)$ denotes that domain element x has property P in the output structure if it has property Q in the input structure.

(15) These types of logical interpretations can be used to define phonological input-output maps such as **a** $\rightarrow$ **b/c_d** (Strother-Garcia 2019, Chandlee and Jardine 2021, Nelson 2024, Heinz et al. Forthcoming).

a. $$\begin{aligned} \phi_a(x) &:= a(x) \wedge \neg\exists y, z[y \lhd x \lhd z \wedge c(y) \wedge d(z)] \\ \phi_b(x) &:= b(x) \vee (a(x) \wedge \exists y, z[y \lhd x \lhd z \wedge c(y) \wedge d(z)]) \\ \phi_c(x) &:= c(x) \\ \phi_d(x) &:= d(x) \end{aligned}$$

b. 

[2]These also require a domain formula, copy set, and licensing function which I will ignore for the purpose of this talk.

(16) The model-theoretic approach also provides a way for identifying substructures with logic. This is useful when thinking about static phonotactic or morpheme structure constraints.

a. Does structure A contain the substring cad?

b. $\varphi := \exists x[\exists y[y \lhd x \wedge c(y)] \wedge \exists z[x \lhd z \wedge d(z)] \wedge a(x) \wedge \texttt{TRUE}]$

c. The last conjunct isn't strictly necessary, but will make more sense as we progress through the handout.

(17) These can be thought of as boolean functions over strings, but we can generalize these to be functions from strings to some member of a set S.

a. $f : \Sigma^* \rightarrow \{\texttt{TRUE}, \texttt{FALSE}\}$

b. $f : \Sigma^* \rightarrow S$

Weighted Logic

(18) The types of set S we are interested in are called **semirings** (Mohri 1997, Goodman 1999, Droste and Gastin 2009, Roark and Sproat 2007, Chandlee 2017, Heinz et al. Forthcoming). Here are some of their properties:[3]

a. A binary "addition" operator $\oplus$

b. A binary "multiplication" operator $\otimes$

c. A value **0** which is an identity for $\oplus$

d. A value **1** which is an identity for $\otimes$

(19) The boolean semiring contains the values $\mathbb{B} = \{\texttt{TRUE}, \texttt{FALSE}\}$.

a. Disjunction acts as addition with FALSE as its identity $([b \in \mathbb{B}] \vee \texttt{FALSE}) = b)$.

b. Conjunction acts as multiplication with TRUE as its identity $([b \in \mathbb{B}] \wedge \texttt{TRUE}) = b)$.

(20) Both the probability semiring ($\mathbb{R}_{\geq 0}$) and the natural number ($\mathbb{N}$) semirings have addition and multiplication as their operators and 0 and 1 as their identities.

(21) **Weighted MSO Logic** adds the following conditions to standard MSO logic.

a. MSO logic (roughly): quantification over individuals and sets of individuals plus all standard logical connectives and definitions.

b. $s \in$ S is an atomic formula

c. Negation is only allowed for $\phi \in \{P(x), x \lhd y, x \in X\}$

d. $\phi \wedge \psi$ is interpreted as $\phi \otimes \psi$

e. $\phi \vee \psi$ is interpreted as $\phi \oplus \psi$

f. $\forall x \phi$ is interpreted as $\phi(x_1) \otimes \phi(x_2) \otimes \cdots \otimes \phi(x_n) \;\; \forall x \in \mathcal{D}$

[3] See Appendix A for more details.

g. $\exists x\phi$ is interpreted as $\phi(x_1) \oplus \phi(x_2) \oplus \cdots \oplus \phi(x_n) \;\; \forall x \in \mathcal{D}$

(22) Weighted logic allows us to use what are essentially the same formula, but make different types of claims about a structure (*cf.* Mayer 2025). Suppose we had a string $abab$.

a. $\exists x, y[a(x) \wedge x \lhd y \wedge b(y) \wedge \texttt{TRUE}]]$

b. $[a(1) \otimes (1 \lhd 1) \otimes b(1) \wedge s] \oplus [a(1) \otimes (1 \lhd 2) \otimes b(2) \wedge s] \oplus \ldots \oplus [a(4) \otimes (4 \lhd 4) \otimes b(4) \wedge s]$

c. If we use the boolean semiring we turn all the $\oplus$ symbols back to disjunction and all the $\otimes$ symbols back to conjunction. Assuming s once again is TRUE, this returns a truth value which tells us whether or not the string contains the sequence ab ***(binary wellformedness)***.

d. If we use the natural numbers semiring we turn all the $\oplus$ symbols into regular addition and all the $\otimes$ symbols into regular multiplication. Assuming s equals 1, this returns a count of how many times a string contains the sequence ab ***(counting violations)***.

e. If we use the natural numbers semiring we turn all the $\oplus$ symbols into regular addition and all the $\otimes$ symbols into regular multiplication. Assuming s equals an arbitrary value, this returns a weight based on how many times a string contains the sequence ab ***(weighted violation/continuous wellformedness)***.

(23) What will be key for the proposed analysis is the **finite language semiring** (*cf.* Beros and Higuera 2014).

a. The set S is the set of all finite languages (stringsets).

b. $\oplus$ is union ($\cup$), $\otimes$ is language concatenation ($\cdot$), **0** is the empty set ($\emptyset$), and **1** is the set containing the empty string ($\{\lambda\}$). Importantly, $\emptyset \cdot A = A \cdot \emptyset = \emptyset$.

c. When this is used, we can define functions from strings to *sets of strings* ($f : \Sigma^* \rightarrow \mathcal{P}(\Sigma^*)$)

d. We can now write logical formula that compute optional **b→c/a_a**.

e. $\forall x \exists y, z[\overbrace{(a(x) \wedge \{a\}) \vee (b(x) \wedge \{b\})}^{\text{a's and b's are "faithful"}} \vee \overbrace{((b(x) \wedge a(y) \wedge a(z) \wedge y \lhd x \lhd z) \wedge \{c\})}^{\text{b's between two a's are changed to c's}}]$

f. The extension of this function is: {(aba,{aba,aca}), (aca, {aca}), (ba,{ba}), (ababa, {ababa, acaba, abaca, acaca})…}

(24) Here is a semi-worked through example for post-nasal voicing:

a. A general formula where the boxes are temporary placeholders:

$$\varphi(x) := (a(x) \wedge \square) \vee (d(x) \wedge \square) \vee (t(x) \wedge \square) \vee (n(x) \wedge \square) \vee (t(x) \wedge n(p(x)) \wedge \square)$$

b. Recall from above that we can switch out the conjunction and disjunction symbols for $\otimes$ and $\oplus$ symbols which are then interpreted in relation to a specific semiring:

$$\varphi : (x) = (a(x) \otimes \square) \oplus (d(x) \otimes \square) \oplus (t(x) \otimes \square) \oplus (n(x) \otimes \square) \oplus (t(x) \otimes n(p(x)) \otimes \square)$$

c. If we wanted to use this formula to count how many times there was an instance of a voiceless stop after a nasal we can change the $\otimes$ and $\oplus$ symbols into regular $\times$ and $+$ while also filling the boxes with 0 for each subformula corresponding to a unigram, and 1 to the specific bigram formula that is only true when there is a voiceless stop after a nasal.

$$\varphi(x) := (a(x) \times 0) + (d(x) \times 0) + (t(x) \times 0) + (n(x) \times 0) + (t(x) \times n(p(x)) \times 1)$$

d. Suppose we want to evaluate the string $a_1 n_2 t_3 a_4$. We evaluate each index individually. Recall also that for this semiring $\texttt{TRUE} = 1$ and $\texttt{FALSE} = 0$.

e. First we check domain element 1...

$$(a(1) \times 0) + (d(1) \times 0) + (t(1) \times 0) + (n(1) \times 0) + (t(1) \times n(p(1)) \times 1)$$

$$(\top \times 0) + (\bot \times 0) + (\bot \times 0) + (\bot \times 0) + (\bot \times \bot \times 1)$$

$$(1 \times 0) + (0 \times 0) + (0 \times 0) + (0 \times 0) + (0 \times 0 \times 1)$$

$$(0) + (0) + (0) + (0) + (0)$$

$$0$$

f. Then for domain element 2...

$$(a(2) \times 0) + (d(2) \times 0) + (t(2) \times 0) + (n(2) \times 0) + (t(2) \times n(p(2)) \times 1)$$

$$(\bot \times 0) + (\bot \times 0) + (\bot \times 0) + (\top \times 0) + (\bot \times \bot \times 1)$$

$$(0 \times 0) + (0 \times 0) + (0 \times 0) + (1 \times 0) + (0 \times 0 \times 1)$$

$$(0) + (0) + (0) + (0) + (0)$$

$$0$$

g. And for domain element 3...

$$(a(3) \times 0) + (d(3) \times 0) + (t(3) \times 0) + (n(3) \times 0) + (t(3) \times n(p(3)) \times 1)$$

$$(\bot \times 0) + (\bot \times 0) + (\top \times 0) + (\bot \times 0) + (\top \times \top \times 1)$$

$$(0 \times 0) + (0 \times 0) + (1 \times 0) + (0 \times 0) + (1 \times 1 \times 1)$$

$$(0) + (0) + (0) + (0) + (1)$$

$$1$$

h. And so on. Since we're counting, we also want to add everything up. So ultimately we want to wrap our formula in an existential quantifier because this "adds" up the results of each index.

i. What if instead, we wanted to use the finite language semiring? This will result in us mapping a single string to potentially multiple output strings.

j. We update φ to look like this:

$$\varphi(x) := (a(x) \cdot \{a\}) \cup (d(x) \cdot \{b\}) \cup (t(x) \cdot \{t\}) \cup (n(x) \cdot \{n\}) \cup (t(x) \cdot n(p(x)) \cdot \{d\})$$

k. We play the same game, checking domain element 1...

$$(a(1) \cdot \{a\}) \cup (d(1) \cdot \{d\}) \cup (t(1) \cdot \{t\}) \cup (n(1) \cdot \{n\}) \cup (t(1) \cdot n(p(1)) \cdot \{d\})$$

$$(\top \cdot \{a\}) \cup (\bot \cdot \{d\}) \cup (\bot \cdot \{t\}) \cup (\bot \cdot \{n\}) \cup (\bot \cdot \bot \cdot \{d\})$$

$$(\{\lambda\} \cdot \{a\}) \cup (\emptyset \cdot \{d\}) \cup (\emptyset \cdot \{t\}) \cup (\emptyset \cdot \{n\}) \cup (\emptyset \cdot \emptyset \cdot \{d\})$$

$$(\{a\}) \cup (\emptyset) \cup (\emptyset) \cup (\emptyset) \cup (\emptyset)$$

$$\{a\}$$

l. Then we check domain element 2...

$$(a(2) \cdot \{a\}) \cup (d(2) \cdot \{d\}) \cup (t(2) \cdot \{t\}) \cup (n(2) \cdot \{n\}) \cup (t(2) \cdot n(p(2)) \cdot \{d\})$$

$$(\bot \cdot \{a\}) \cup (\bot \cdot \{d\}) \cup (\bot \cdot \{t\}) \cup (\top \cdot \{n\}) \cup (\bot \cdot \bot \cdot \{d\})$$

$$(\emptyset \cdot \{a\}) \cup (\emptyset \cdot \{d\}) \cup (\emptyset \cdot \{t\}) \cup (\{\lambda\} \cdot \{n\}) \cup (\emptyset \cdot \emptyset \cdot \{d\})$$

$$(\emptyset) \cup (\emptyset) \cup (\emptyset) \cup (\{n\}) \cup (\emptyset)$$

$$\{n\}$$

m. Domain element 3 is where it gets interesting...

$$(a(3) \cdot \{a\}) \cup (d(3) \cdot \{d\}) \cup (t(3) \cdot \{t\}) \cup (n(3) \cdot \{n\}) \cup (t(3) \cdot n(p(3)) \cdot \{d\})$$

$$(\bot \cdot \{a\}) \cup (\bot \cdot \{d\}) \cup (\top \cdot \{t\}) \cup (\bot \cdot \{n\}) \cup (\top \cdot \top \cdot \{d\})$$

$$(\emptyset \cdot \{a\}) \cup (\emptyset \cdot \{d\}) \cup (\{\lambda\} \cdot \{t\}) \cup (\emptyset \cdot \{n\}) \cup (\{\lambda\} \cdot \{\lambda\} \cdot \{d\})$$

$$(\emptyset) \cup (\emptyset) \cup (\{t\}) \cup (\emptyset) \cup (\{d\})$$

$$\{t, d\}$$

n. In position three, the output set now contains two symbols: t for the "faithful" mapping and d corresponding to a post-nasal voicing rule. For this analysis, we also want to wrap our formula in the universal quantifier because we want to "multiply" aka do language concatenation across each domain element. Nonetheless, we can see that the core formula is exactly the same for counting as it is for mapping to multiple strings.

Phonological Knowledge and Phonological Usage

(25) The general idea for the system I am proposing is to split the computation into two functions: one for phonological **k**nowledge (competence) which determines possible outputs and one for phonological **u**sage which determines a specific output at a given moment (performance).

a. Knowledge/Competence generates possible outputs by mapping input strings to output sets of strings using the finite language semiring ($k : \Sigma^* \to \mathcal{P}(\Sigma^*)$).

b. Usage/Performance generates weights for the set of possible outputs using the probability semiring ($u : \mathcal{P}(\Sigma^*) \to \mathbb{R}^n_{\geq 0}$) and chooses an output with something like `argmax` ($x \in \mathcal{P}(\Sigma) \mid \texttt{argmax}(u) = x$).

c. Crucially, this only works if the factors that determine the weights include at least some dynamic properties. This seems to be the exact desired result if we honor the spirit of competence/performance. Knowledge is static(ish) but performance is dependent on environmental conditions.

Nasal Place Assimilation

(26) In English, coronal nasals assimilate in place to a following obstruent across word boundaries. The status of non-coronal nasal assimilation is contested, but it has been argued to be absent in certain (Borowsky 1986, Avery and Rice 1989, Coleman et al. 2016). This gives us pairs like i[m] [p^{h}]ort Jefferson $\sim$ fro[m] [p^{h}]ort Jefferson, i[ŋ] [k^{h}]anada $\sim$ fro[m] [k^{h}]anada, and i[n] [t^{h}]acoma $\sim$ fro[m] [t^{h}]acoma

(27) Production and perception rates vary.

a. Production (Buckeye Corpus; Dilley and Pitt 2007): pre-labial (18.5%) and pre-velar (21.8%)

b. Production (Audio British National Corpus; Coleman et al. 2016): overall (20%)

c. Perception (Coetzee 2016): pre-labial (45%) and pre-velar (35%)

(28) Coetzee (2016) shows that factors like *speech rate*, and *experiment type* are influence the rate of assimilation. We can turn these into functions that give weights to use in our logic.

a. $\texttt{speechRate} : \{\text{slow,faster,fastest}\} \to \mathbb{R}_{\geq 0}$ b. $\texttt{expType} : \{\text{exp1,exp2}\} \to \mathbb{R}_{\geq 0}$

(29) We can first implement the **k** function using the finite language semiring.

a. $\varphi := \forall x[(\mathfrak{ŋ}(x) \wedge \{\mathfrak{ŋ}\}) \vee (\mathfrak{n}(x) \wedge \{\mathfrak{n}\}) \vee (\mathfrak{m}(x) \wedge \{\mathfrak{m}\}) \vee (\mathfrak{n}(x) \wedge \texttt{preLabStop}(x) \wedge \{\mathfrak{m}\}) \vee (\mathfrak{n}(x) \wedge \texttt{preVelStop}(x) \wedge \{\mathfrak{ŋ}\}) \vee (?(x) \wedge \{?\})$

b. Extension: {(nt,{nt}), (np,{np,mp}), (nk,{nk,ŋk}), (mp,{mp})}

(30) We can implement the **u** function using the probability semiring.

a. $\varphi := (\texttt{speechRate} \vee \texttt{expType}) \wedge \forall x[(n(x) \wedge \texttt{preLabStop}(x) \wedge w_1) \vee (n(x) \wedge \texttt{preVelStop}(x) \wedge w_2)]$

b. This gives a weight to each output of the k function which can be turned into a probability distribution by using `softmax`. A hypothetical example: $\{np, mp\} \rightarrow [4.3, 17.11] \leftrightarrow [0.21, 0.79]$.

c. Previous work (e.g., Coetzee 2016) has suggested that extragrammatical factors act as offsets to grammatical (phonological) factors governing preference for certain forms. This is captured in the equation above. But there are lots of other options we can explore.

d. Something like `expType` is probably not the actual factor governing selection and it is therefore an empirical endeavor to explore the cognitive underpinnings of these things.

e. Furthermore, many interacting factors can lead to the selection of one variant over another due to what Bayley (2013) refers to as "the principle of multiple causes".

Conclusion

(31) a. Composing the functions suggests they may be viewed as a single function, but only the latter is affected by extra-grammatical factors. This plus the choice of semiring echo the competence/performance distinction.

$$\overbrace{\Sigma^* \rightarrow \mathcal{P}}^{\substack{\text{Generation:}\\ \text{Competence}}} \overbrace{(\Sigma^*) \rightarrow \langle \Sigma^*, \mathbb{R}^n_{\geq 0} \rangle \rightarrow \Sigma^*}^{\substack{\text{Weighting/Selection:}\\ \text{Performance}}}$$

b. The law of frequency matching becomes a second order effect. Variation is rarely "free" and is regularly explained by internal and external factors. If these create probability distributions, then behavior would match the observed probabilities if language users correctly learn which factors affect choice.

c. Formal logic provides a way to be explicit about the systems one is analyzing and how certain commitments such as discrete phonological knowledge and modularity can coexist with gradient usage data.

Epilogue

(32) a. Since this approach only alters the outputs (i.e., semantic interpretation), all the algebraic results (Lambert 2022, Lambert and Heinz 2023, 2024) about functions should still hold since these are all based on input structure (i.e., syntax). In that sense, this approach does not increase complexity. Optionality does not break subregularity.

b. One of the advantages of the model-theoretic approach is the flexibility in representations it offers. This kind of disappears with the weighted logic approach sketched here. One possibility would be to define feature-level string formulas (i.e., feature transducers) and then combine them together somehow *a la*

Markowska and Heinz (2026). I would appreciate any suggestions people have here about how to integrate other types of representations.

c. With only domain knowledge, semi-deterministic functions are not identifiable in the limit. If you have both domain knowledge and an oracle which answers, "is y a valid output of x" these types of functions are identifiable in the limit (Beros and Higuera 2014).

d. Full MSO Weighted Logic is probably not necessary for our purposes. I only used FO in the examples today. Idsardi (2025) suggests that some type of the two-variable guarded fragment of FO is sufficient for phonological analysis.

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A Part 1: Monoids and Semirings for Phonological Transductions

(33) a. Monoids and semirings are algebraic structures.

b. They are of interest to computational linguists because they allow for different output interpretations (i.e., *semantics*) of the same input structure (i.e., *syntax*).

c. Mohri (1997) showed how weighted finite state transducers (which require a specific semiring for interpretation) were useful for language and speech processing tasks.

d. Goodman (1999) shortly after showed how different outputs of syntactic parsing could be described by the same algorithm by substituting the operations of different semirings.

e. Roark and Sproat (2007) discuss semirings in their *Computational approaches to Morphology and Syntax* book

f. Gorman and Sproat (2021) discuss how weighted finite state transducers form the backbone of their Python package `pynini` which is a wrapper for the more general `openfst` package.

g. Chandlee and Heinz (2017) suggest the use of semirings to solve the "fuzzy membership" problem in phonotactics.

h. Mayer (2025) uses monoids to discuss phonotactic grammars and ultimately argues that "constraints and representations in the grammar can be studied independently of the values the grammar assigns," and "[t]his flexibility allows us to engage with a broader range of empirical phenomena for which categorical or gradient models provide better approximations while still relating these phenomena to the same core linguistic knowledge."

(34) a. A monoid is a set K which contains an associative binary operator $\circ$ and an identity element e.

b. A semiring is a set S with two binary operators $\oplus$ and $\otimes$ and two identity elements 0 and 1 which act as identities.

(35) Semirings are therefore the combination of two monoids (sometimes also called a *dioid*) with some additional stipulations:

a. $\oplus$ must be commutative: $a \oplus b = b \oplus a$.

b. 0 must act as an annihilator for $\otimes$: $0 \otimes a = 0 = a \otimes 0$.

c. $\otimes$ must right distribute over $\oplus$: $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$.

(36) a. In a standard finite-state transducer, strings built from an alphabet Σ are written to an output alphabet Δ.

b. The machine reads a single symbol at a time and writes a corresponding output string which are concatenated together as it moves from beginning to end.

c. With *deterministic* weighted finite-state transducers (Mohri 1997), the set of possible outputs ($\Delta^\star$) is replaced with the monoid set K and concatenation is replaced with a corresponding binary operation over that set $\circ$.

d. Here are some common monoids and semirings:

Name	K/S	$\circ$	e	$\oplus$	$\otimes$	0	1
String	$\Delta^\star$	$\cdot$	λ				
Boolean	$\{\texttt{TRUE}, \texttt{FALSE}\}$	$\wedge$	TRUE	$\vee$	$\wedge$	FALSE	TRUE
Natural	$\mathbb{N}$	$+$	0	$+$	$\times$	0	1
Real Interval	$[0, 1]$	$\times$	1	$+$	$\times$	0	1
FIN	$\{L \subseteq \Sigma^\star \mid \text{L is finite}\}$	$\cdot$	$\{\lambda\}$	$\cup$	$\cdot$	$\emptyset$	$\{\lambda\}$

(37) a. Monoids are used for deterministic transducers while semirings are used for non-deterministic transducers.

b. "Multiplication" defines how to interpret each path and "addition" defines how to combine different paths together. In the monoid case, the single operator acts like "multiplication" in that it simply defines how to interpret the path.

(38) a. To give an intuition of how this all works, consider a generic postnasal voicing process over the input alphabet $\Sigma = \{$a,p,b$\}$. The transducer computing this function maps input strings like *ampa* to *amba*, but also *amba* to *amba*. This corresponds to the leftmost transducer in Figure 1.

b. The center transducer in Figure 1 takes the same structure, but makes the outputs empty. The numbers in this instance are simply used for indexing/reference.

(39) The table below gives possible values for the empty outputs for the machine above:

	Output Label											Mappings	
Monoid	1	2	3	4	5	6	7	8	9	10	11	*ampa*	*amba*
String	λ	λ	a	p	b	m	λ	m	a	b	b	*amba*	*amba*
Boolean	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\bot$	$\top$	$\bot$	$\top$
Natural	0	0	0	0	0	0	0	0	0	1	0	1	0
Real	1	0.2	0.2	0.2	0.2	0.2	0.2375	0.2375	0.2375	0.05	0.2375	0.0004	0.0019
FIN	$\{\lambda\}$	$\{\lambda\}$	{a}	{p}	{b}	{m}	$\{\lambda\}$	{m}	{a}	{b, p}	{b}	{*amba, ampa*}	{*amba*}

(40) a. The different monoids provide a way to use the same machine to compute different types of functions that phonologists have been interested in. State 10 ends up being the crucial state because this is where a *p* following an *m* has been read on the input string.

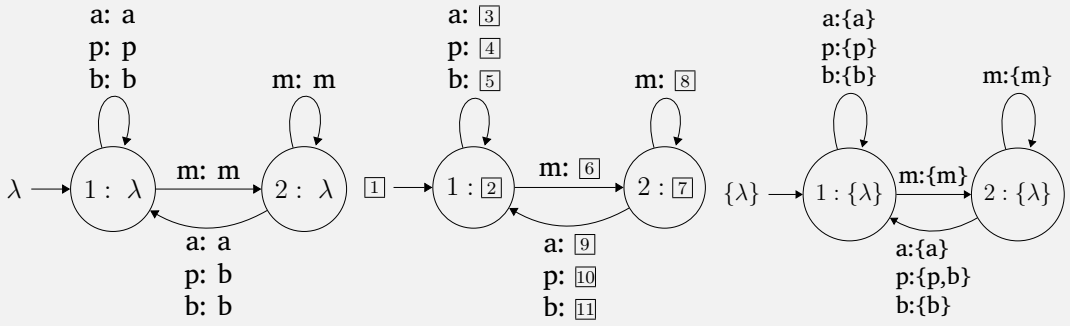


Figure 1: Deterministic single output function for post-nasal voicing (left), output-empty version (center), and semi-deterministic multiple version (right).

b. The string monoid computes functions from strings to strings: *p* is changed to a *b* on the output to "repair" the illicit structure.

c. The Boolean monoid computes functions from strings to booleans which is akin to binary wellformedness: any instance of an *mp* sequence would make the form illformed

d. The Natural monoid computes functions from strings to natural numbers which can be thought of as counting the number of violations of a given markedness constraint in a string: each instance of *mp* adds 1 to the output.

e. The Real interval monoid computes functions from strings to values between 0 and 1 which extends binary wellformedness to gradient wellformedness: strings with an *mp* sequence receive a smaller weight than strings which do not contain the sequence.

(41) a. This brings us to the finite language monoid (FIN) which provides a way to understand optional processes as still being deterministic.

b. The binary operation operating over FIN in this case is *language concatenation* which is defined as $XY = \{xy \mid x \in X, y \in Y\}$.

c. To account for optional post-nasal voicing, it requires changing each transition from single strings to sets of strings. Most of the transitions will be singleton sets containing the string that was read on the input.

d. The only distinction is that now the transition from state 2 to state 1 after reading "p" will output the set of strings {p,b}.

e. This will result in two strings that will persist throughout the rest of the computation, one where post-nasal voicing didn't apply and one where it did. This is shown in the rightmost transducer above.

f. The final output will therefore be a set of size 2^n where n is equal to the number of times post-nasal voicing could have applied in the string.

B Recent Approaches to Optionality

(42) There are three general approaches with optimization-based grammars: Stochastic OT and Noisy Harmonic Grammar assume that Harmonic Evaluation is a *noisy* procedure which can lead to local re-ranking if the constraint weights are close enough together (Boersma 1998, Boersma and Hayes 2001).[4]

(43) a. In stochastic modeling, noise is often represented using a normal distribution (bell curve) centered around 0.

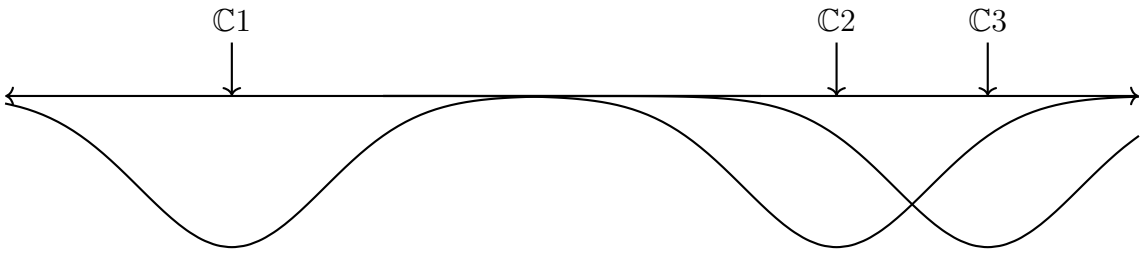
b. This "noise" is then added to a deterministic value to make the entire system non-deterministic or *stochastic*.

c. A probability distribution emerges indirectly in the long term.

(44) a. The figure below shows potential constraint rankings and their corresponding "noise" distributions.

b. Since $\mathbb{C}1$ is ranked sufficiently higher than the other constraints it will always remain the highest ranked.

c. On the other hand $\mathbb{C}2$ and $\mathbb{C}3$ are close enough to where their local ranking can flip on different evaluations.



(45) a. Another approach to optionality/variation uses MaxEnt grammars (Hayes and Wilson 2008, Hayes 2022).

b. MaxEnt does not have a noisy evaluation procedure. Instead, a normalization procedure is performed over the output candidate weights.[5]

c. This is a three step process where first a harmony score is calculated for each candidate, this is used to determine an eHarmony score, these second scores are the basis for normalization.

d. Below shows how the probability distribution (rightmost column) for some

[4]There is a variant of NHG where noise is instead added to each cell rather than to the constraint weight (Goldrick and Daland 2009). Interestingly, this was proposed specifically to account for speech errors which sometimes yield non-grammatical forms: "Being blind to the nature of constraints, the disruptions caused by random noise will not be constrained to produce only those mappings that can specified by the grammar."

[5]MaxEnt grammars are usually *phonotactic grammars*, but they can be used for input-output mappings as well.

syllable types are determined based of a ONSET constraint with weight 3 and NOCODA constraint with weight 2.

x	*#V	*C#	Score	P*(x)	P(x)
C	0	2	2	0.1353353	0.1019709
V	3	0	3	0.04978707	0.03751299
CV	0	0	0	1	0.7534685
VC	3	2	5	0.006737947	0.00507683
CVC	0	2	2	0.1353353	0.1019709

The final approach is a "competing grammars" approach exemplified by Anttila (1997). He gives the following two explanations:

1. A candidate is predicted by the grammar if it wins in some tableau.
2. If a candidate wins in n tableaux and t is the total number of tableaux, then the candidate's probability of occurrence is n/t.

Given three constraints $\mathbb{C}1$, $\mathbb{C}2$, $\mathbb{C}3$, there are $3!$ ways in which they can be ordered. This gives us six grammars/tableaux. The idea behind this third approach is that the probability of one form over another is tied to the proportion of rankings that choose that candidate as the winner. Suppose we have two candidates x_1 and x_2. x_1 violateds $\mathbb{C}1$ and $\mathbb{C}2$ while x_2 violates $\mathbb{C}3$. The following table shows the six possible rankings and which candidate would win in each case.

Ranking	Winner
$\mathbb{C}1 \gg \mathbb{C}2 \gg \mathbb{C}3$	x_2
$\mathbb{C}1 \gg \mathbb{C}3 \gg \mathbb{C}2$	x_2
$\mathbb{C}2 \gg \mathbb{C}1 \gg \mathbb{C}3$	x_2
$\mathbb{C}2 \gg \mathbb{C}3 \gg \mathbb{C}1$	x_2
$\mathbb{C}3 \gg \mathbb{C}1 \gg \mathbb{C}2$	x_1
$\mathbb{C}3 \gg \mathbb{C}2 \gg \mathbb{C}1$	x_1

This predicts that x_2 should appear about two-thirds of the time while x_1 should appear about one-third of the time. A probability distribution emerges from the types of forms the constraint set predicts.

C A Different Approach: Optionality as a property of the phonetic implementation

(46) a. Under the BLUEPRINT MODEL OF PRODUCTION Nelson and Heinz (2026), a recent model of the phonetics-phonology interface, optionality is explained based on properties of the post-phonological phonetic production.

b. The BLUEPRINT MODEL OF PRODUCTION, explicitly structures the phonetic production module such that phonological knowledge (surface form), lexical knowledge (underlying form), and extra-grammatical knowledge all interact to produce a phonetic output.

c. Nelson and Heinz (2026) show that access to both the underlying and surface form allows for straightforward explanations of phenomena such as incomplete neutralization and variation in homophone duration while maintaining discrete phonological knowledge.

(47) Under this approach phonological knowledge itself is never optional: it exists as one type of information that is consulted during language use. Optionality therefore shifts from being a property of the grammar to choosing to apply phonological knowledge when computing the phonetic exponent a given lexical item (Nelson 2025).

(48) a. The basic story with the BLUEPRINT MODEL OF PRODUCTION is that you can account for subtle phonetic variation by considering both surface and underlying form.

b. We implement this with a scaling formula for a given phonetic cue c.

c. To account for optionality, I add a parameter $v \in \{0, 1\}$ that chooses whether or not to use the surface form (phonological knowledge) in the computation.

d. When $v = 1$ no phonological knowledge is applied and when $v = 0$ phonological knowledge is applied.

e. The scaling formula is:

$$c = c_{UR} \times \textbf{max}(v, i^{\alpha}) + c_{SR} \times (1 - \textbf{max}(v, i^{\alpha}))$$

(49) a. Global optionality in Waoro is reported by Osborn (1966).

b. Labial voicing is a context free process which voices labial stops.

c. Examples (50–51) provide specific surface forms where there is free variation between the all or nothing cases (a) but the cases where there is a mismatch in application are not observed (b).

(50) *weak*

a. [paroparera] ∼ [**b**aro**b**arera]

b. * [paro**b**arera] ∼ * [**b**aroparera]

(51) *he will put them*

a. [apaupute] ∼ [a**b**au**b**ute]

b. * [apau**b**ute] ∼ * [a**b**aupute]

(52) The figure below shows the abstract characterization of how optionality emerges. The center box with a function labeled A is what implements the scaling formula above. P corresponds with SR, L corresponds with UR, and E corresponds with v.

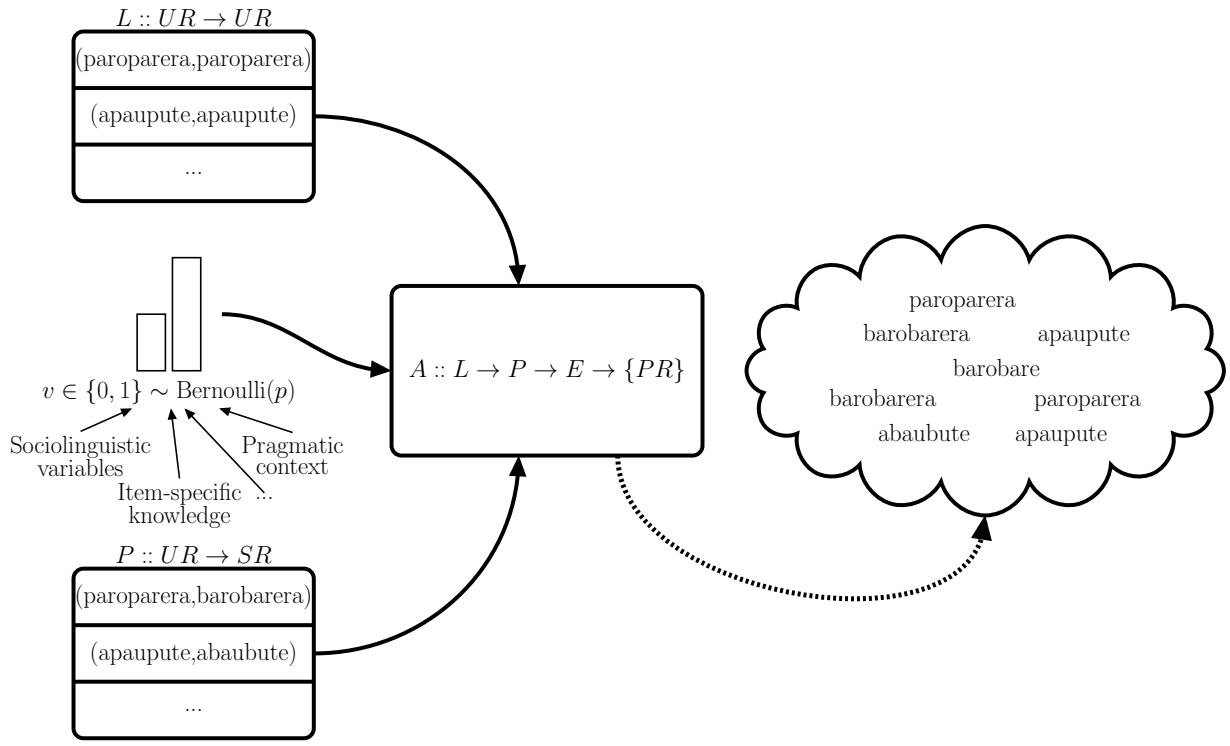


Figure 2: An example of data generation under the Blueprint Model of Production for labial voicing in Waoro. The lexicon and phonology are discrete, but the parameter v is controlled by various external factors which create a probability distribution over its application.

(53) a. As Lakatos (1970) writes, "A 'model' is a set of initial conditions (possibly together with some of the observational theories) which one knows is *bound* to be replaced during the further development of the programme, and one even knows, more or less, how."

b. The Blueprint Model of Production was originally proposed to provide an explanation for how discrete phonological knowledge could still account for continuous phonetic phenomena such as incomplete neutralization and variation in homophone duration. Here I have shown how a slight alteration of some of the original assumptions could lead to a further understanding of optionality and variability as it pertains to rate of application of phonological knowledge.